

Jia Huang

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Research Focus

I study how to make generative video models usable as world models: keeping them physically consistent and improving them through preference and reward post-training. My work covers simulator-grounded video data, reward-guided post-training of video world models, and region-aware preference optimization for video inpainting.

Interests: video world models; video generation and editing; preference and reward optimization for video diffusion; 3D/4D vision; embodied perception.

Education

Columbia University Jul. 2025 – Oct. 2026
M.S. in Applied Analytics · GPA 3.89/4.00 New York, NY

University of Liverpool & Xi'an Jiaotong-Liverpool University Sep. 2021 – Jul. 2025
B.Eng. (Hons) in Data Science and Big Data Technology (dual degree) · First Class Honours Suzhou, China

GRE General: Quantitative 170/170, Verbal 158/170

Publications and Manuscripts

Video generation and multimodal retrieval

- [1] **Jia Huang**, Hongyou Zhou, Qi Huang, Hongbin Liu, Zhengqi Sun, Lumin Xu. “PVI-DPO: Perceptual Video Inpainting via Informative-Safe Preference Mining and Region-Aware Direct Preference Optimization.” *Under review at AAAI 2027 (advanced to Phase 2)*.
- [2] Peng Zhang, Hongyou Zhou, **Jia Huang**. “SVP-Rerank: Difficulty-Conditioned Speech–Visual–Procedural Evidence Reranking for Medical Instructional Video Retrieval.” *NLPCC 2026 Shared Tasks & Evaluation (accepted)*. Ranked 1st of 10 teams on Track 2.

Other manuscripts (causal inference)

- [3] Pengyun Wang, Qi Huang, **Jia Huang**, Yutong Zhou. “Robust Propagation-Integrated Inference for Observational Studies with Network Interference.” *Under review at ICLR 2027*.
- [4] Pengyun Wang, Qi Huang, **Jia Huang**, Yutong Zhou. “Simultaneous Inference for Proximal Causal Learning with Weak Proxies.” *Under review at ICLR 2027*.

Research Experience

University of Rochester — VIGIL, ARPA-H PARADIGM Program Jun. 2026 – Present
Research Collaborator (remote) · Primary Developer, Voice Intelligence Rochester, NY

Advisor: Prof. Chenliang Xu

- Lead the Rochester voice-intelligence workstream of VIGIL, a multi-institution clinical-assistance system, covering data collection, modeling, evaluation, ROS 2 integration, and deployment.
- Designed a two-stage “Hey VIGIL” trigger (openWakeWord GRU front end plus a verifier head on the frozen Qwen3-ASR-1.7B audio encoder), raising participant-disjoint F1 from 0.75 to 0.97.
- Integrated streaming Qwen3-ASR transcription and voice authorization into the live ROS 2 (Humble) agent loop on the University of Michigan pilot workstation, routing requests to state control, visual QA, and medical QA with Piper TTS responses.
- Built the data-collection platform (React/FastAPI/Supabase; 1,673 recordings from 39 participants), participant-disjoint cross-validation and hard-negative regression tests, and released one-command inference bundles on Hugging Face.

- Studied failure modes of large video world models in long-horizon generation: background drift, object deformation, camera-trajectory mismatch, and inconsistency when regions are re-observed.
- Built a simulator-grounded corpus of ~5,000 moving-camera physical-interaction sequences (drop, collision, roll, containment) with aligned RGB, depth, object masks, camera poses, intrinsics, and event-aware prompts.
- Post-trained LingBot-Fast with LoRA adaptation and GRPO-based reward guidance, using rewards and diagnostics built from RAFT optical flow, DINOv2 features, epipolar/Sampson geometry, and V-JEPA 2/VideoREPA representations to score camera adherence, object identity, temporal consistency, and physical plausibility.

PVI-DPO: Preference Optimization for Video Inpainting

Lead Researcher and First Author · Independent project, with collaborators incl. Lumin Xu (Zoom)

2025 – 2026

- Formulated a region-aware, winner-safe DPO objective that weights masked, boundary, and visible-context regions separately, with anchor and gap regularizers that widen preference margins without degrading high-quality winners.
- Mined 2,949 medium-difficulty preference pairs from YouTube-VOS by scoring eight stochastic rollouts per clip with 22 spatial, perceptual, and temporal quality signals.
- Built reproducible training and evaluation pipelines on DAVIS and YouTube-VOS: comparison with eight baselines, ablations, localized, perceptual, and temporal metrics, and a human preference study.

Industry Experience

Shanghai Mingpin Medical Data Technology — Re Research Institute

Medical Vision Research Intern

2025

Shanghai, China

- Developed lesion and organ segmentation for multi-center imaging (nnU-Net, Swin UNETR), MedSAM-based clinician-in-the-loop pre-annotation, and conditional latent diffusion for rare-lesion synthesis under limited data.

iFLYTEK Co., Ltd.

Robotics Vision Intern

Apr. 2024 – Aug. 2024

China

- Built machine-vision prototypes and validated CNN/YOLO-based perception models with R&D teams.

Robotics Systems Projects

Maritime Search-and-Rescue Surface Vessel; Urban Intelligent Rescue Vehicle

Research lead (vessel, 2023–24); vision-module lead (vehicle, 2022–23)

2022 – 2024

XJTLU

- Deployed open-vocabulary perception (Grounding DINO, SAM), multi-object tracking (ByteTrack), visual-inertial SLAM (ORB-SLAM3, RTAB-Map), and A*/Hybrid-A* planning on ROS/ROS 2.

Honors and Awards

- 1st of 10 teams, NLPCC 2026 Shared Task, Track 2 (medical instructional video retrieval) 2026
- First Class Honours, University of Liverpool 2025
- National Second Prize (Finals) and Provincial First Prize, 19th iFLYTEK Intelligent Vehicle Competition 2024

Technical Skills

Generative models	PyTorch, Diffusers; LoRA fine-tuning; DPO and GRPO post-training; multi-GPU BF16 training on SLURM; W&B
Vision and 3D/4D	DINOv2, V-JEPA 2, RAFT; SAM, Grounding DINO, ByteTrack; ORB-SLAM3, RTAB-Map; epipolar geometry
Systems	Python, C + +, SQL, Bash; Git, Docker, ROS/ROS 2, FastAPI
Languages	Chinese (native), English (fluent)